EXP-2

EXPT NO: 2 OCC AND LOAD CHARACTERISTICS OF D.C SHUNT GENERATOR

Aim:

To conduct load test on separately excited generators and to obtain the characteristics

Apparatus Required:

S .no	Apparatus	Range	Type	Quantity
1	Volt meter	(0-300)V	MC	1
2	Ammeter	(0-2.5)A	MC	1
3	Ammeter	(0-5)A	MC	1
4	Rheostat		Wire wouned	1
5	Rheostat		Wire wouned	1
6	Connecting wires	-	-	As required

Fuse rating calculation for field and armature:

No load test

10 % of rated current (full load current)

Load test

125 % of rated current (full load current)

Precautions

1. Motor side field rheostat should be kept at minimum resistance position.
2. Generator side field rheostat should be kept at maximum resistance position.
3. Starter should be in off position before switching on the supply.
4. The DPST switch must be kept open.Procedure for open circuit test
5. Procedure
6. Connections are given as per the circuit diagram.
7. The motor is started with the help of THREE POINT starter.
8. Adjust the motor speed to rated speed by adjusting motor field rheostat when the generator is disconnected from the load by DPST switch 2.
9. By varying the generator field rheostat gradually, the open circuit voltage [Eo] and corresponding field current (If) are tabulated up to 125 % of rated voltage of generator.

10. The motor is switched off by using DPST switch 1 after bringing all the rheostats to initial position.

Procedure for Load test:

1. Connections are given as per the circuit diagram
2. The prime mover is started with the help of three point starter and it is made to run at rated speed when the generator is disconnected from the load by DPST switch 2.
3. By varying the generator field rheostat gradually, the rated voltage [Eg] is obtained.
4. The ammeter and voltmeter readings are observed at no load condition.
5. The ammeter and voltmeter readings are observed for different loads up to the rated current by closing the DPST switch 2.
6. After tabulating all the readings the load is brought to its initial position.
7. The motor is switched off by using DPST switch 1 after bringing all the rheostats to initial position.

Tabulation for Open Circuit Test:

OBSERVATION TABLE

S.No.	Voltage (V)	Current (A)
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.21
5	133	0.23
6	135	0.24

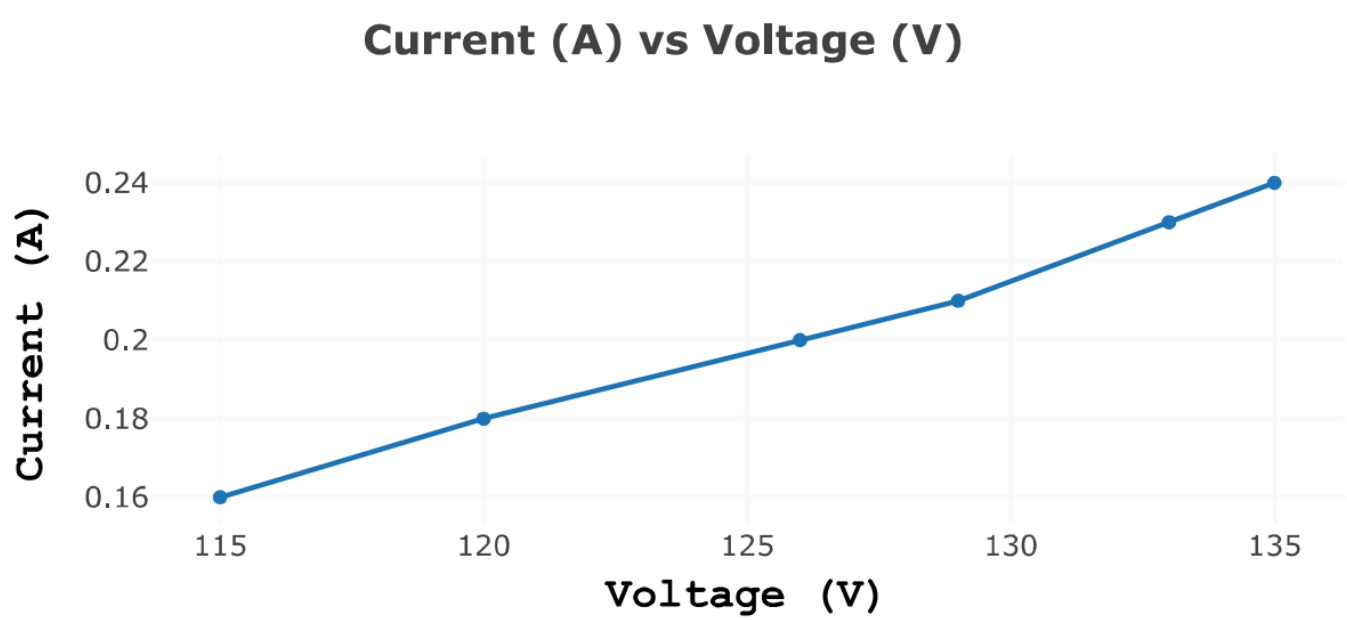
Tabulation for Load Test:

Observation Table

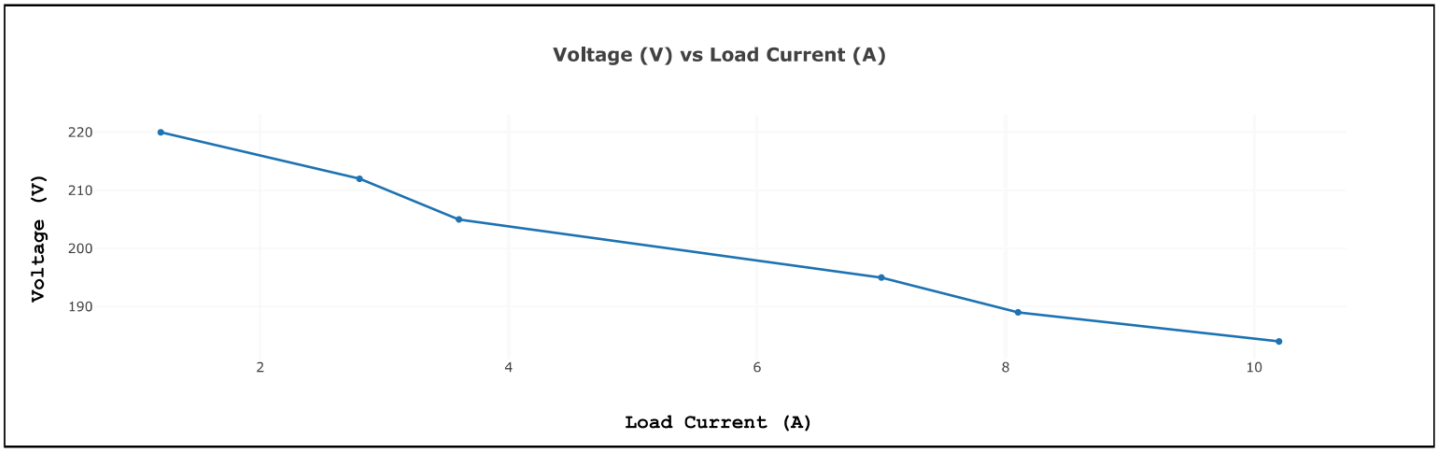
S.No.	Current (A)	Voltage (V)
1	1.2	220
2	2.8	212
3	3.6	205
4	7	195
5	8.1	189
6	10.2	184

Model graph:

Open circuit graph:



Load circuit graph:



Calculation:

Open circuit calculation:

CALCULATION FOR OPEN CIRCUIT CHARACTERISTICS

formula for voltage to current ratio.

$$R = \frac{E_0}{I_f}$$

Reading 1-6:-

$$R_1 = \frac{115}{0.16} = 718.75$$

$$R_2 = \frac{120}{0.18} = 666.66$$

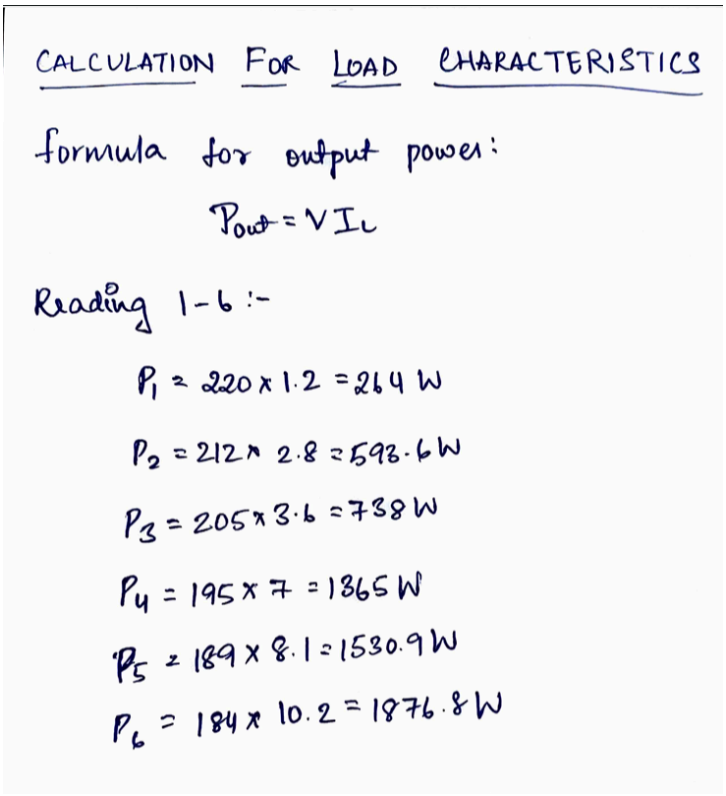
$$R_3 = \frac{126}{0.20} = 630$$

$$R_4 = \frac{129}{0.21} = 614.28$$

$$R_5 = \frac{133}{0.23} = 578.26$$

$$R_6 = \frac{135}{0.24} = 562.5$$

Load circuit calculation:



Open circuit characteristics:

The curve of generated EMF V against field current I_f at constant rated speed, with the load disconnected.

Load Characteristics:

Terminal voltage V_t against load current I_L at constant rated speed, with the field rheostat fixed after setting rated voltage at no load.

Result:

The load test on separately excited generators and to obtain the characteristics was verified.